Learning conditional language in continuous image space

Imagized Language Model Project

Written language includes forms that cannot be faithfully represented by a fixed character inventory. We study language modelling with visible writing as both the input and generated output, using continuous internal states and a single consumer GPU. A causal reader trained on Chinese raster streams ranks the next glyph at 19.97% top-1 on 2,048 development windows, compared with 12.26% for a bigram control and 18.36% after shuffling earlier history. Realizing that predictive information as readable output remains difficult. A separately trained glyph diffusion prior and noisy visual critic retain 21.88% completed-image ranking, yet recover only one of 256 target identities after four-sample selection. We therefore test a directly conditional four-glyph velocity denoiser with 22.18 million total parameters. Its single-GPU run completes 8,000 updates in 361.6 seconds. Direct first-position identity is 1.17%, and block-mean identity is 0.586%, with an ordered-minus-shuffled gain of 0.098 percentage points. These measurements separate visual prediction, glyph synthesis and conditional language generation, and define the remaining requirements for an independent model that learns and writes language through images.

Written language is encountered as marks on paper, screens and historical surfaces. Its visual form carries information about script, handwriting and material history. Modern character encodings provide a powerful interface to many such marks, but a finite inventory cannot express every damaged, transitional or unencoded form. An image-based language model could retain that evidence throughout reading and generation, with transcription offered as an optional output view.

Pixel-based language modelling already has substantial precedents. PIXAR demonstrates autoregressive language generation in pixel space and identifies the difficulty of generating readable text[1]. Continuous autoregression can use a diffusion loss in place of vector quantization[2], while Diffusion Forcing models continuous sequences under independently chosen noise levels[3]. These results motivate a visual language interface, but do not establish that a small glyph generator acquires linguistic knowledge or that image input automatically reduces training cost.

We investigate the conditional generation problem directly. The model receives raster history, predicts a distribution of future writing, emits pixels and rereads those pixels. Chinese provides a demanding initial case: stroke configurations must remain coherent while context selects among many similar written forms. The present study uses one canonical font to isolate this coupling before introducing handwriting or historical scripts.

Results

A continuous visual interface preserves the written signal

Each observed cell is a 32×32 ink image. An orthonormal transform maps its signed pixels into a 1,024-dimensional continuous field. The transform is invertible and contains no learned character dictionary. An ordered sequence of these fields enters a causal transformer. The deployed model receives no Unicode values, OCR transcripts, character IDs or vocabulary logits (Fig. 1). Host-side rendering uses encoded source text during data preparation. This distinction matters: the current benchmark studies learning from rendered Chinese, rather than discovering a script from raw archaeological photographs.

Ordered image history contains next-writing information

The strongest established reader result uses 2,048 deterministic development windows and an evaluator-only bank of 1,024 canonical glyph images. Full-history next-image top-1 is 19.97%, above the image unigram at 1.42%, the symbolic bigram at 12.26% and the prefix-shuffled condition at 18.36%. The ordered history improves target log probability by 0.174 nat over the shuffled control. This result supports bounded conditional prediction from image sequences.

These figures measure selection among completed images. The evaluator bank does not enter the model and cannot act as a deployed writer. Thresholding the predicted continuous field can mix strokes from several plausible futures. For squared-error prediction, the population optimum is $f^*(H) = \mathbb{E}[X \mid H]$. A conditional mean need not be a valid glyph when several futures are possible. Sharp copying, target ranking and sampled language generation therefore require separate measurements.

A glyph prior does not automatically bind to language

To separate visual synthesis from contextual prediction, we train an unconditional glyph prior and a noisy visual compatibility critic while freezing the causal reader. The combined runtime contains 23.27 million parameters. The critic scores an arbitrary noisy image against the visible history. In the ideal density-ratio limit,

$$\nabla_x \log p_t(x \mid H) = \nabla_x \log p_t(x) + \nabla_x \log \frac{p_t(x \mid H)}{p_t(x)}. \qquad (1)$$

The construction is related to classifier-guided diffusion[4].

On 512 development windows, the clean critic reaches 21.875% top-1, versus 20.898% with shuffled history. Its first guided sample reaches zero target identities in 256 cases. Selecting the highest clean critic score among four samples reaches 1/256, with the same result after shuffling history. The generated glyph-like marks are diverse and nonblank, yet fail conditional identity (Fig. 2).

This experiment identifies a reader–writer gap without proving a unique cause. At the terminal corruption level, target signal power is 2.43×10^{-7}, and an exact conditional guidance term should vanish. That limit also occurs in successful conditional diffusion. It cannot alone explain the failure. Insufficient conditional-score accuracy, off-manifold gradients, limited prior quality and finite sampling remain compatible explanations. We consequently test a directly conditional writer rather than interpret clean-image ranking as evidence of a generative solution.

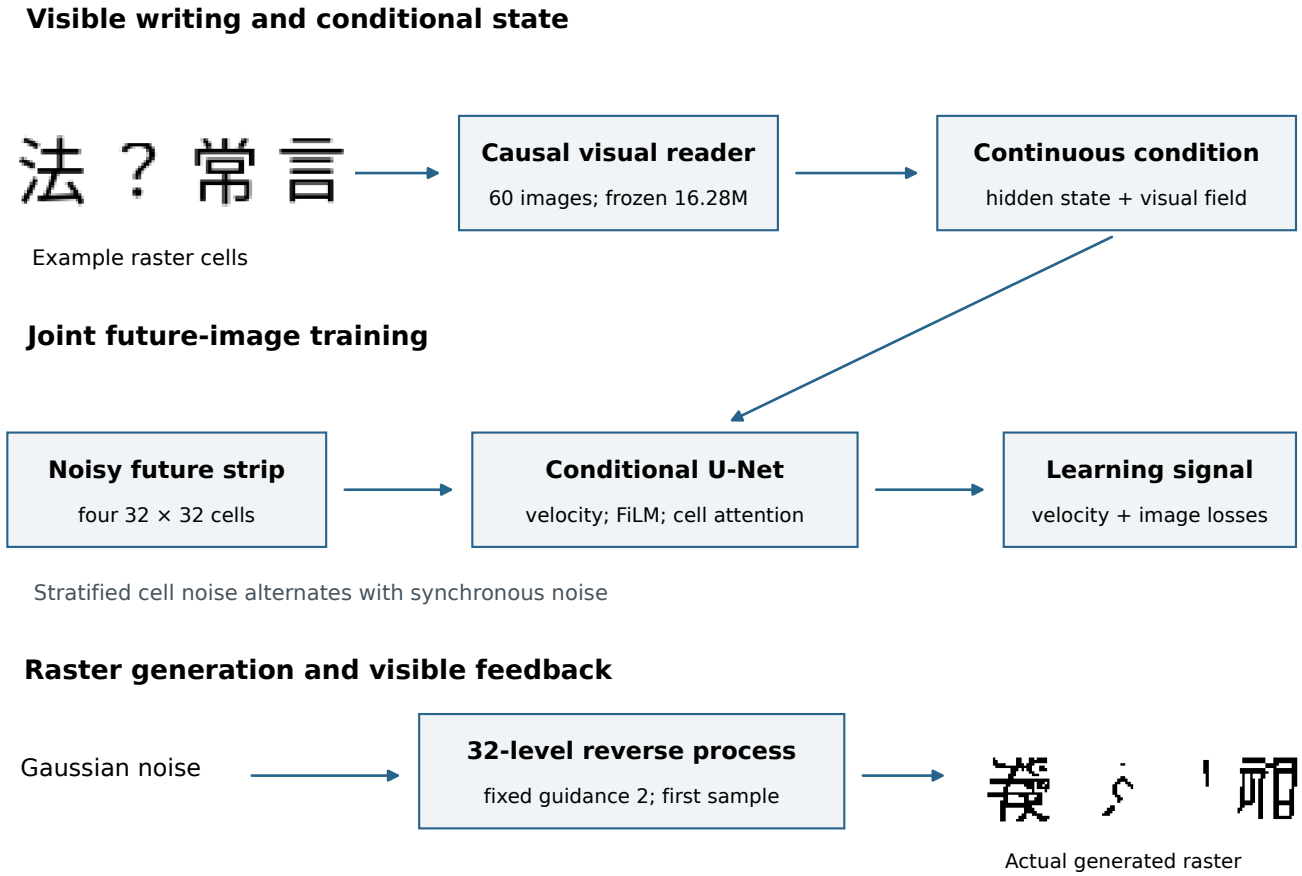


Fig. 1 | A visual language interface and its conditional generation test. a, Rendered or native writing supplies ordered image cells to a causal visual reader. Text rendering is an optional host interface. **b**, A conditional denoiser uses the frozen reader state and a noisy four-cell strip to predict diffusion velocity. Future cells exchange information through convolutions and pooled-cell attention. **c**, Sampling starts from noise, emits four visible cells and appends their pixels to the next context. Optional transcription follows generation and is absent from the model. The schematic shows the implemented experimental contract, without claiming that its generated content is already correct.

Direct conditional block generation tests the missing coupling

The new writer models a joint four-cell future conditioned on 60 visible cells. It reuses the 16.28M-parameter reader and fine-tunes a 5.90M-parameter strip denoiser. A projected reader state modulates each residual scale, and a four-cell attention layer exchanges future information at the bottleneck. There is no separate guidance critic or candidate selector.

For signed clean writing x_0, noise ε and cumulative signal $\bar{\alpha}_t$, the training target is velocity:

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\varepsilon, \tag{2}$$

$$v^* = \sqrt{\bar{\alpha}_t}\varepsilon - \sqrt{1-\bar{\alpha}_t}x_0. \tag{3}$$

The clean estimate is $\hat{x}_0 = \sqrt{\bar{\alpha}_t}x_t - \sqrt{1-\bar{\alpha}_t}\hat{v}$. Training alternates independently stratified future noise levels with synchronous levels that cover the sampling distribution. Fixed visual-field and foreground losses retain the writing signal. Intermediate target-field alignment follows the motivation of representation alignment[5], using a fixed raster transform rather than a semantic teacher.

The run completes all 8,000 updates in 361.6 s, using 0.543 GiB peak allocated CUDA memory. Evaluation uses one primary sample from 256 prefixes under paired initial noise. Ordered first-position identity is 1.17%, compared with 1.17% after shuffling and 0.00% with a zero condition. Across all four future cells, identity is 0.586%, with an ordered-minus-shuffled difference of 0.098 percentage points and a record-bootstrap 95% interval of [0.000, 0.307]. Pixel F1 is 0.320. The registered complete-mechanism decision is `reject`. The raw image panel shows the first fixed examples without reranking or selection (Fig. 2).

A post-result motor diagnostic supplies the intended first-cell image to the same trained writer on 64 fixed prefixes. Exact plan and image-field supply raise pixel F1 from 0.378 to 0.459 and target identity from 1/64 to 5/64. The fixed inverse transform reproduces the supplied raster with F1 1.0. Because this test supplies the answer, it provides no autonomous-language evidence. It shows that the current learned writer still loses substantial shape information even with an exact visual condition.

Discussion

The results separate a useful visual predictive state from a useful visual language generator. The established causal reader learns context-dependent next-writing information, while subsequent writers can erase that information or model the marginal appearance of glyphs. A complete visual language model must bind meaning and sequence to the exact marks it generates.

An image interface enlarges the representable output space, including the possibility of forms absent from a character encoding. It does not remove the need to learn syntax, knowledge or long-range dependencies. This study has not demonstrated general instruction following, etymology answers, handwriting

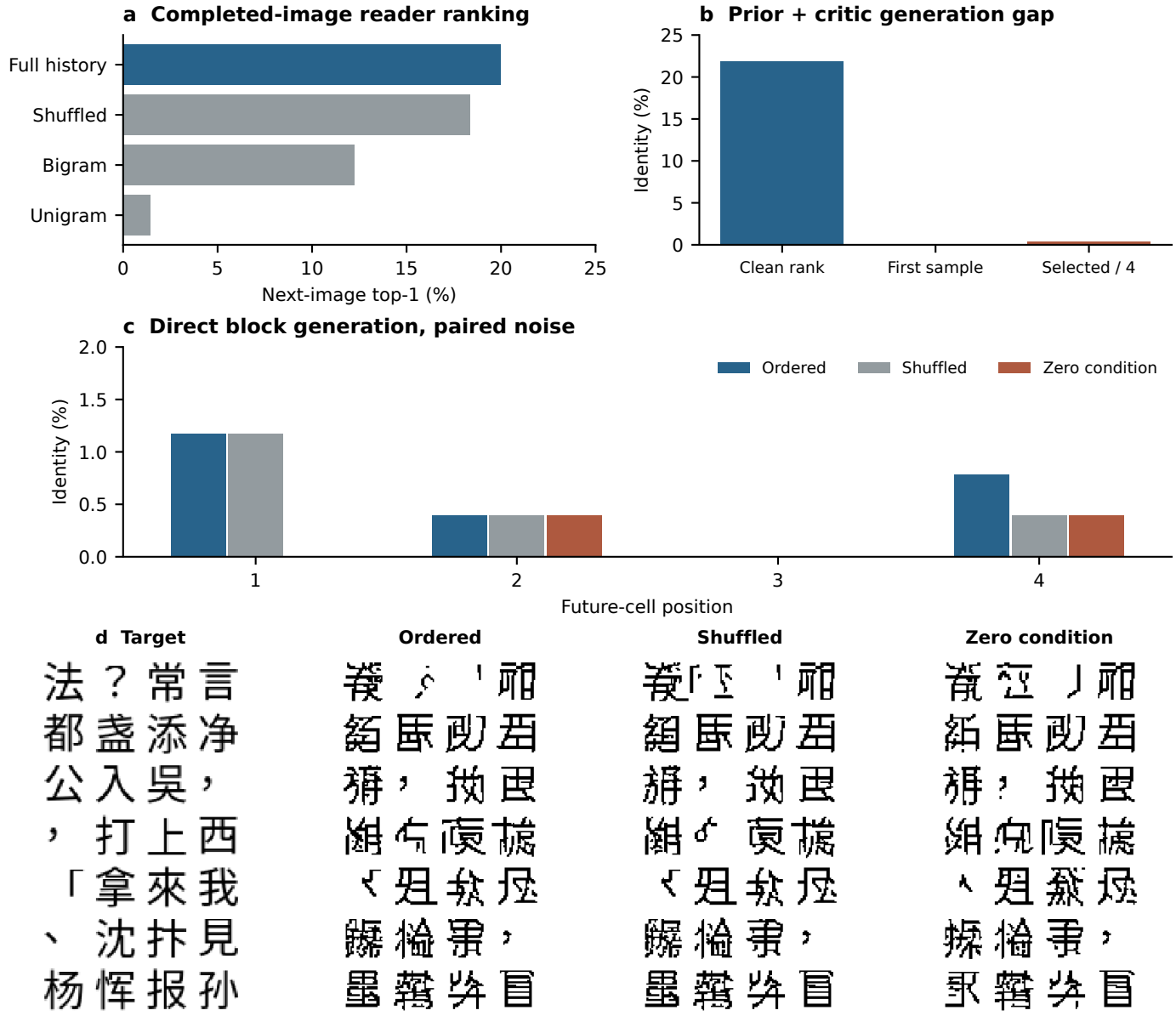


Fig. 2 | Prediction and generated-image evidence. a, The established causal reader ranks completed next images above unigram, bigram and shuffled-history controls on 2,048 windows. These are ranking measurements, not generated text. **b**, Prior-plus-critic guidance retains completed-image ranking but almost never generates the target identity. **c**, Direct conditional block generation reports all four positions under ordered, shuffled and zero conditions. Bars use the same starting noise and single primary samples. **d**, Fixed, unselected examples show the target and all generated controls. Images are raw model outputs with nearest-neighbour enlargement. No OCR or image-bank replacement is applied.

transfer or arbitrary historical-script generation. It also lacks independent training-seed replication and a capability-matched text-model comparison. Small allocated memory in a bounded experiment establishes feasibility on one GPU, rather than a general efficiency advantage over language models.

The next decisive tests should first require exact visual copying on unseen forms, then conditional continuation and short answers with counterfactual prompts. High-noise denoising must improve with the correct history, and the same improvement must survive autonomous sampling and visible feedback. Broader books and offline teacher-generated exercises can supply later training data while leaving the final inference path independent. A larger corpus or longer training is warranted only when these measurements show that the conditional mechanism learns the intended task.

Methods

Data, rendering and partitioning

The public Chinese stream contains 7,017 provenance-tracked records. Records are assigned deterministically to training, development or a sealed partition before rendering. Original and simplified views inherit their source split. The new block experiment uses one fixed Noto Sans CJK font, 26-pixel writing inside 32-pixel cells, without image augmentation. A 65-cell training segment supplies the first 60 cells as context and cells 61–64 as future writing. The remaining cell is unused. No string, source identifier or symbolic target enters a model method. Sealed records remain unopened.

Training and numerical verification

The reader is frozen. Copied glyph-prior weights initialize the denoiser, which is then optimized as a velocity predictor. AdamW uses a learning rate of 5×10^{-5} for transferred layers and 2×10^{-4} for new modules, 400 warmup updates, cosine decay to 10%, batch size eight, BF16 arithmetic and gradient clipping at one. EMA decay is 0.999. Checkpoints retain model, optimizer, RNG, dataset offset and provenance hashes. Interrupted and continuous smoke runs produce byte-identical model and EMA weights.

The field objective applies an orthonormal transform directly to floating signed predictions. Thresholding is confined to output and identity evaluation. This distinction is verified by an isolated pixel-gradient test. The primary velocity MSE is supplemented by foreground, Dice, field and representation losses at weight 0.10 each, plus an ordered-history margin at weight 0.25. Supplementary Information gives the complete objective and gates.

Sampling, controls and statistics

Sampling uses 32 registered DDIM schedule levels and classifier-free scale two. Four future cells start from Gaussian noise. The same noise is reused for ordered, suffix-preserving shuffled and zero-condition histories. Only one sample contributes to the primary block result. Generated cells are appended as visible rasters for a second four-cell block.

An evaluator-only canonical bank measures nearest-raster identity after generation. Targets outside the bank receive no identity credit, and bank coverage is reported per position. This measurement can assign a nearest identity to an illegible pseudo-glyph, so direct images and surface statistics accompany every result. Pixel F1 alone is also insufficient because many Chinese glyphs share strokes. No OCR is used by the model or the reported identity evaluator.

Confidence intervals use 2,000 bootstrap resamples of source records, retaining all windows from a record together. Paired differences use identical windows and initial noise. Intervals describe development-record variability for this trained checkpoint, not variability across training seeds. The current development partition has informed a sequence of experiments and is not a final independent generalization test.

Data and code availability

Implementation, protocols, evidence summaries and scripts are available in `https://github.com/lachlanchen/ImagizedLanguageModel`. Manifest and checkpoint hashes identify the exact local research artifacts. Restricted reference books are not redistributed. The complete research record and Supplementary Information distinguish earlier foundation provenance from the present raster-only deployment experiment.

References

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